

Blockchain Technology

Section 1: Fundamentals of Blockchain

Q1. What is Blockchain technology and why is it considered reliable?

- **Answer:** Blockchain is a decentralized and distributed digital ledger technology that records transactions across many computers securely and transparently. It is highly reliable and tamper-proof because once data is recorded, it cannot be altered.

Q2. Describe the structure of a block in a Blockchain.

- **Answer:** A blockchain consists of a series of connected blocks. Each block contains:
 - **Data:** Details regarding the transactions.
 - **Timestamp:** The specific time the transaction occurred.
 - **Hash:** A unique digital signature identifying the block.
 - **Previous Hash:** A link that connects the current block to the one before it.
- **Note:** This structure ensures security; if data is changed, the hash changes, which breaks the chain.

Q3. Explain the step-by-step working process of a Blockchain.

- **Answer:** The process follows these six steps:
 1. A user initiates a transaction.
 2. The transaction is broadcast to a network of nodes (computers).
 3. Nodes use algorithms to validate the transaction.
 4. Verified transactions are grouped into a block.
 5. That block is added to the existing blockchain.
 6. The transaction is completed and becomes a permanent record.

Q4. What are the key features of Blockchain?

- **Answer:**
 - **Decentralization:** No central authority is in control.
 - **Transparency:** All participants can see the transactions.

- **Security:** Data is protected via cryptography.
 - **Immutability:** Once data is added, it cannot be modified.
 - **Consensus Mechanism:** Nodes must agree before any data is added.
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Section 2: Types and Applications

Q5. List and define the different types of Blockchains.

- **Answer:**
 - **Public Blockchain:** Open for everyone to join (e.g., Bitcoin).
 - **Private Blockchain:** Controlled strictly by a single organization.
 - **Consortium Blockchain:** Managed by a specific group of organizations.
 - **Hybrid Blockchain:** A mix of both public and private blockchain features.

Q6. What are the pros and cons of using Blockchain?

- **Answer:**
 - **Advantages:** High security , transparency and trust , reduction in fraud , elimination of middlemen , and faster transaction speeds.
 - **Disadvantages:** High energy consumption , scalability issues , technological complexity , and a lack of clear regulations.
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Section 3: Real-World Applications

Q7. How is Blockchain applied in Cryptocurrency?

- **Answer:** Cryptocurrency is digital money (like Bitcoin or Ethereum) that operates without a central bank. Transactions are verified by a network using cryptography and recorded permanently on the blockchain.
- **Uses:** Online payments , international transfers , and trading.
- **Benefits:** Low costs , no intermediaries , and fast processing.

Q8. How does Blockchain improve Supply Chain Management?

- **Answer:** It tracks the movement of goods from production to the final delivery. Every step of the journey is recorded as unalterable information shared between manufacturers, suppliers, and retailers.
- **Uses:** Tracking food from farm to consumer , verifying if a product is authentic , and managing logistics.
- **Benefits:** Reduces errors and fraud , improves transparency , and increases overall efficiency.